

ACADEMIC SUMMARY NOTE

“AI for Americans First”

*American AI Protectionism, Reshaping of the Global Technological Order,
and Consequences for France and Europe (2026–2030)*

Fabrice Pizzi

Université Paris Sorbonne, 2026

Academic Summary Note, February 2026

Abstract

This study analyses the mechanisms and consequences of American AI protectionism under the Trump 2.0 administration, integrating four dimensions that are typically treated separately in the literature: energy, semiconductors, compute, and regulation. Drawing on an empirical diagnosis for the period 2020–2026, the construction of a Compute, Adjusted Competitiveness Index (CACI), and a 2×2 scenario matrix, the research demonstrates that the combination of tariffs (25%, Section 232) and export controls creates a measurable structural competitive advantage (US/EU CACI ratio of 7 to 12:1), paradoxically accelerates the construction of an alternative Chinese AI ecosystem, and fragments the global technological order into competing blocs. A comparative analysis of regional responses (Europe, South America, Asia) reveals fundamentally differentiated dependency trajectories. A prospective addendum formalises a fifth cross, cutting scenario, the “Grand Decoupling” of 2028, based on American Cloud Sovereignty Mandates and the dissociation between the physical factor and the sovereign factor in AI compute ($C(r) = C_phys(r) \times F_sov(r)$), revealing that France and Europe conflate physically installed capacity with operationally sovereign compute. For France, the study identifies a strategic action window for 2026–2028 and recommends a targeted strategic autonomy grounded in the nuclear advantage, the Mistral AI champion, and the European regulatory framework.

Keywords: artificial intelligence, technological protectionism, semiconductors, export controls, sovereign compute, AI geopolitics, France, United States, China, Cloud Sovereignty Mandates, Grand Decoupling, sovereign factor F_sov , CLOUD Act, CADA, sovereignty washing, jurisdictional AI blocs

1. Subject and Research Question

Artificial intelligence has established itself since 2023 as the primary vector of economic innovation and geopolitical competition. The AI value chain displays unprecedented concentration: the United States controls 74% of global AI compute; five American hyperscalers (Microsoft, Amazon, Alphabet, Meta, Oracle) are projected to invest between 660 and 690 billion dollars in capital expenditure for 2026 alone; and Nvidia holds approximately 80% of the AI accelerator market.¹

In this context, the Trump 2.0 administration has transformed the export controls initiated under Biden (2022–2025) into a hybrid protectionist regime combining tariffs and regulatory restrictions. This study poses the following question: to what extent does American AI

protectionism create a measurable structural competitive advantage, and what are the differentiated consequences for France, Europe, and other regions of the world?

The originality of this research lies in the integration of four dimensions that are typically siloed in the literature: energy trajectories of data centers, the semiconductor market, the distribution of AI compute, and the American regulatory timeline. No economic study had previously formalised an integrated “Trump 2.0, AI protectionism” scenario; this research fills that gap.

2. Methodological Framework

The methodology rests on three pillars. First, a longitudinal empirical diagnosis (2020–2026) based on data from the IEA, McKinsey, SIA, Epoch AI, and regulatory sources (BIS, White House, European Parliament). Three critical curves are reconstructed: data center energy consumption by region (TWh), the semiconductor market (value, AI share), and the distribution of AI compute (IT load in GW, FLOPs by region).

Second, the construction of a Compute, Adjusted Competitiveness Index (CACI), a composite index integrating four variables: peak available FLOPs (weighted 40%), energy cost per FLOPs (25%), AI human capital (20%), and regulatory access to compute (15%). The CACI enables quantification of structural asymmetry between regions and projection of divergence trajectories.²

Third, a 2×2 scenario matrix crossing two axes (intensity of US protectionism: moderate/strong; European response: passive/active) to generate four scenarios for 2026–2030, each calibrated on the CACI and tested against three empirical tipping points (EU energy saturation 2028, China foundry consolidation 2029, alternative chip maturity 2030). A prospective addendum extends the CACI model through a decomposition of the compute factor: $C(r) = C_{\text{phys}}(r) \times F_{\text{sov}}(r)$, where $F_{\text{sov}}(r)$ measures the fraction of regional compute operating outside US jurisdiction ($0 \leq F_{\text{sov}} \leq 1$). This extension reveals that the $C(r)$ variable used in scenarios A–D measures physically installed capacity, not operationally sovereign capacity — two metrics that diverge considerably in regions dominated by American hyperscalers.

3. Main Findings

3.1 A Three, Tier Protectionism

The analysis identifies a three, level cumulative protectionist architecture. The first tier consists of export controls (inherited from Biden, transformed by Trump), which segment the world into three tiers: unlimited access for 20 close allies (Tier 1), quantitative caps for the rest (Tier 2), prohibition for adversaries (Tier 3). The second tier, an innovation specific to Trump, consists of 25% tariffs (Section 232, 15 January 2026) on advanced AI semiconductors, creating a direct cost differential between American companies (exempt) and non, American ones.³ The third tier is the capitalist gravitational effect: 660–690 billion dollars in annual capex, plus Japanese (550 billion) and Emirati investments converging toward American soil, self, reinforcing compute concentration without additional regulatory intervention.

3.2 Measured Competitive Advantage

The CACI quantifies a US/EU ratio of 7:1 to 12:1, reflecting a compute gap (US: 75 GW, EU: 35 GW), a FLOPs cost differential (US \$0.5/TFlop vs. EU \$1.2–1.8/TFlop), and an AI productivity gap (+30% US, +12% EU in AI, intensive sectors). Four transmission mechanisms

are identified: training cost asymmetry (GPT, 4: \$100M US vs. \$500M EU), cloud concentration (72–80% of European AI workloads on US hyperscalers), the productivity gap, and innovation rent capture (scale effects + network effects).⁷

3.3 Systemic Paradoxical Effects

The study demonstrates that protectionism produces three paradoxical effects. First, the restrictions accelerate the construction of an autonomous Chinese AI ecosystem (Huawei Ascend 910c, DeepSeek, V3, \$125+ billion in investments in 2025) rather than neutralising it. Second, Tier 2 countries (Brazil, India, ASEAN) are pushed toward Chinese technology partnerships (ByteDance: \$38 billion in Brazil, \$8.8 billion in Thailand), creating a global technology bifurcation. Third, Tier 1 allies (Japan, South Korea) co-finance US supremacy (Japan: \$550 billion invested on American soil) rather than building their own autonomy. The result is not a unipolar order but a world fragmented into technological blocs.⁵

3.4 The 2028 Grand Decoupling: A Fifth Cross, Cutting Scenario

A prospective addendum models a qualitative shift distinct from the four initial scenarios: the “Cloud, Nationality” Pivot. Whereas scenarios A–D analyse degrees of protectionism within a framework of physical control over semiconductors, this scenario models a change in nature — a shift from control through flows (export controls on chips) to control through the jurisdictional layer (sovereign operation of existing clusters). Its legal, technical foundations are already operational in 2026: the CLOUD Act (2018) establishes that an H100 cluster physically installed in Ireland or Dubai remains legally American¹⁰; the American AI Action Plan of 23 July 2025 introduces location verification features embedded in advanced AI chips, the first technical substrate for remote throttling.¹¹

The central paradox of the scenario lies in the dissociation between the physical factor and the sovereign factor. Applying the formula $C(r) = C_{\text{phys}}(r) \times F_{\text{sov}}(r)$ to 2025 data reveals that the jurisdictions that have invested most in attracting American hyperscaler data centers are precisely those whose CACI collapses most severely under Cloud Sovereignty Mandates: the United Arab Emirates ($F_{\text{sov}} \approx 0.12$, collapse 60–80%), Singapore ($F_{\text{sov}} \approx 0.18$, 55–75%), and counterintuitively the EU ($F_{\text{sov}} \approx 0.28$, 30–50%), while China ($F_{\text{sov}} \approx 0.98$) remains unaffected.¹² This result demonstrates that the current European policy of attracting data centers conflates installed compute with sovereign compute, a confusion that the CADA (Cloud and AI Development Act, proposed by the Commission in Q1 2026) is only beginning to formalise.¹³ Scenario B (Fracture + Mandates) produces a double scissors effect — scarce chips and conditional compute simultaneously — pushing the CACI US/EU ratio potentially beyond 40:1. Scenario C (Gigafactories under European jurisdiction) is the only one to absorb the shock by structurally increasing F_{sov} toward 0.45–0.55.

4. Comparative Regional Analysis

The study conducts a differentiated analysis of the impact of AI protectionism on five regions, revealing structurally distinct dependency trajectories.

Region	Tier	Main Dynamics	Strategic Asset	Main Risk
France / Europe	1	GPU+cloud US dependency (72–80%); InvestAI 200B€ response	Nuclear (70%), Mistral, ASML, AI Act	Geopolitical vendor lock, in; marginalisation if US, Asia bloc closes

Region	Tier	Main Dynamics	Strategic Asset	Main Risk
Brazil / S. America	2	US, China competition ground; TikTok/Scala mega, projects	83% renewable energy mix; dynamic fintech market	Triple fracture (North, South, East, West, intra, regional)
Japan / Korea / Taiwan	1	US co, investment (\$550B); production transfer to US	HBM (SK Hynix), TSMC 90% advanced chips, materials	Asymmetric partnership; erosion of Taiwan silicon shield
India	2	South global pivot; \$200B+ ambition; 'compute export' strategy	1.4B population, tech talent, zero, tax cloud policy	Structural gap (1.4 GW vs 53.7 US); Tier 2 GPU caps
China	3	Forced self, sufficiency; Huawei/DeepSeek alternative ecosystem	Domestic market 1.4B; \$125B+/yr investments	2–3 generation GPU lag; technological isolation

The analysis reveals that geopolitical position (Tier 1/2/3), energy endowment, and proximity to value chains determine trajectories that cannot be reduced to a single model. France benefits from its Tier 1 status but suffers from a structural compute gap. Brazil, classified Tier 2, is the direct theatre of US, China rivalry. Japan, the most integrated ally, co, finances American supremacy. India attempts a “third way.” China is building a parallel ecosystem.

5. Strategic Recommendations for France

The study formulates recommendations structured around five axes and three time horizons.

Axis 1 — Compute infrastructure. Accelerate the 13 European AI Factories (operational by end of 2027), implement Special Compute Zones (fast, track permits, reduced taxation), and deploy the 5 AI Gigafactories under InvestAI (€20 billion). Target: 30–40% of sensitive AI workloads on certified sovereign cloud by 2029. The F_sov distinction requires a reformulation of the success metric: the decisive indicator is not “how many GPUs in Europe” but “how many GPUs under European jurisdiction.”¹⁴ The phenomenon of sovereignty washing — American hyperscalers marketing “sovereign cloud” on European soil while remaining subject to the CLOUD Act — represents the main systemic risk in the current investment attraction policy.⁶

Axis 2 — Nuclear energy. Leverage France’s unique advantage (70% decarbonised nuclear electricity). EDF has identified four industrial sites totalling 2 GW, with the Nuclear for AI initiative (250 MW by end of 2026). Accelerate the 6 EPR 2 reactors (Penly, Bugey, 9,900 MW, construction from 2027), confirm the 8 optional reactors, and support SMRs (NUWARD, Newcleo, Stellaria).⁷

Axis 3 — Technology alliances. Consolidate the ASML, Mistral partnership (€1.3 billion, ASML first shareholder at 11%). Negotiate a second TSMC investment in Europe on advanced nodes. Conclude bilateral EU, Japan and EU, South Korea agreements on supply security (HBM memory, equipment). Build strategic GPU reserves (6–12 months).

Axis 4 — Offensive regulation. Transform the AI Act into a competitive lever: priority for European models in public AI Factories, Brussels effect via mutual recognition agreements, creation of a European “CLOUD Act Shield.”

Axis 5 — Talent. Launch AI scholarships and European talent visas (before end of 2026); guarantee frontier compute access for European researchers (Fluidstack 500,000 GPUs, Mistral Compute, EuroHPC AI Factories).⁸

The critical action window falls between 2026 and 2028: beyond this point, the identified energy and compute tipping point crystallises dependencies.

6. Contributions, Limitations, and Further Research

This research contributes to the literature on five fronts: (i) the analytical integration of trajectories typically siloed (energy, semiconductors, compute, regulation, productivity); (ii) the proposal of the CACI index as a framework for measuring compute, adjusted competitiveness; (iii) the demonstration of the systemic paradoxical effects of AI protectionism (acceleration of the Chinese ecosystem, push of Tier 2 countries toward China, co-financing of US supremacy by allies); (iv) an unprecedented comparative analysis of regional responses revealing structurally distinct dependency trajectories; and (v) the formalisation of the decomposition $C(r) = C_{\text{phys}}(r) \times F_{\text{sov}}(r)$, which reveals the distinction between physically installed compute and operationally sovereign compute — a distinction absent from the existing literature and with direct implications for investment policies in Gigafactories, data centers, and nuclear energy.

Limitations stem from regulatory uncertainty (the BIS final rule of January 2026 could be revised by July 2026), heterogeneity in compute data (the CACI remains an exploratory index), and the time horizon (post-2030 technological disruptions could redistribute advantages). Three further research directions are indicated: empirical calibration of the CACI on survey data, extension to Africa (a continent absent from this study), and dynamic modelling via computable general equilibrium models integrating compute as a factor of production.

7. Conclusion

AI compute is on the verge of becoming the fourth factor of production, structuring access to productivity gains and innovation. The American AI Action Plan of July 2025 now treats the AI stack as an instrument of geopolitical alliance, comparable to the Marshall Plan: access to compute is conditioned on strategic alignment.⁹ Faced with this reshaping, three options present themselves for France: subordinate integration (the Japan model), sovereigntist confrontation (the China model, unrealistic by 2030), or the targeted strategic autonomy this study recommends — sovereignty over segments of comparative advantage (nuclear, ASML, Mistral, AI Act) combined with interoperability with the American ecosystem. The objective is not technological autarky but the capacity for choice. The question is no longer whether the reshaping of the global technological order will occur — it is already underway — but whether we will be its architects or its subjects.

Full Study Structure

Ch.	Title	Pages	Notes
I	Theoretical Framework: Technological Protectionism and AI	12	22
II	Methodology: Scenario Matrix and CACI Index	8	10
III	Empirical Diagnosis 2020–2026	11	20
IV	Mechanisms of US Competitive Advantage	9	19

Ch.	Title	Pages	Notes
V	Prospective Scenarios 2026–2030 and Tipping Points	11	16
VI	Consequences for France and Europe	10	14
VI bis	Consequences for South America and Brazil	11	19
VI ter	Consequences for Asia	12	16
VII	Strategic Recommendations	11	18
Concl.	From AI Protectionism to the Reshaping of the Technological Order	8	3
Total		~120	157

Notes

1 Euronews (February 2026), “Will Big Tech’s AI Spending Crush Europe’s Data Sovereignty?” Capex 2026: Amazon \$200B, Alphabet \$185B, Microsoft \$145B, Meta \$135B, Oracle \$50B. Total: \$660–690B.

2 The CACI index is developed in Chapter II. It draws on “compute, adjusted competitiveness” metrics identified as missing by McKinsey (2024) and the World Economic Forum (2025).

3 Pillsbury Law (January 2026), “Trump Admin Targets Advanced AI Semiconductors.” Section 232: 25% tariff on Nvidia H200, AMD MI325X for re, export. US domestic exemptions. BIS final rule of 15 January 2026.

4 Bruegel (2025), “Why Artificial Intelligence Is Creating Fundamental Challenges for Competition Policy.” Exponential training costs as entry barrier. CFG (2025): EU AI compute = 5% global vs US 75%. McKinsey (2025): productivity +30% US, +12% EU.

5 Carnegie Endowment (May 2025): control/promotion/leverage trilemma. IBTimes India (February 2026): China \$125B+ AI infrastructure 2025. Construction Today (November 2025): Japan \$550B invested in US. Bloomberg/DCD (2025): ByteDance \$38B Brazil (Pecém).

6 European Commission (April 2025), AI Continent Action Plan. 13 AI Factories, InvestAI €200B. CFG (October 2025), “Special Compute Zones: Europe’s Recipe.” Julien Simon, Medium (January 2026), “AI Sovereignty in Europe: A Decision Framework.”

7 World Nuclear News (February 2025): EDF 4 sites, 2 GW. Enki AI (February 2026): EPR 2 (9,900 MW), 20 life, extension reactors (26 GW). Introl Blog (2025): France AI investments €109B, Fluidstack €10B/1 GW.

8 McKinsey (December 2025), “Accelerating Europe’s AI Adoption: The Role of Sovereign AI.” AI scholarships and talent visas to launch before end of 2026. 44% of European tech leaders cite data security as a barrier.

9 CM Trade Law (July 2025), “America’s AI Action Plan.” Pillar III: “export the full AI technology stack to countries willing to join the American AI alliance.” Four principles: allied exports, enforcement, global alignment, protection measures.

10 CLOUD Act (Clarifying Lawful Overseas Use of Data Act), Pub. L. 115, 141 (23 March 2018), Title III, § 103(a), codified at 18 U.S.C. § 2713: “A provider shall comply with the obligations of this chapter (...) regardless of whether such communication, record, or other information is located within or outside of the United States.” Microsoft acknowledged before the Paris Tribunal judiciaire (2024) that it cannot guarantee the sovereignty of European data in the event of a legally grounded injunction in the United States; cited in The Register (22 December 2025).

11 White House, America’s AI Action Plan (23 July 2025), Pillar III, “Strengthen AI Compute Export Control Enforcement.” Michael Kratsios (OSTP Director): “There is discussion about potentially the types of software or physical changes you could make to the chips themselves to do better location, tracking. That is something we explicitly included in the plan.” (APEC Digital and AI Ministerial Meeting, Seoul, August 2025, cited in TechResearchOnline, 5 August 2025). The BIS AI Action Plan tasks the Department of Commerce with identifying these location verification features in collaboration with industry.

12 F_sov estimates calculated from regional cloud market shares: Synergy Research Group (Q3 2025), “Hyperscaler Cloud Infrastructure”; Statista Enterprise Cloud Market Share EU (2025). Share of American hyperscalers (AWS, Azure, GCP): UAE ~88%, Singapore ~82%, EU ~72%, India ~60%, China ~2%. $F_{sov} = 1 - F_{US}(r)$. These estimates constitute an exploratory calibration; rigorous empirical calibration on survey data is identified as a priority research extension.

13 European Commission, Work Programme 2026, Cloud and AI Development Act (CADA), scheduled Q1 2026. Cloud Sovereignty Framework (October 2025): three levels SOV, 1 to SOV, 3, with SOV, 3 requiring full protection against non, European extraterritorial legislation. Cristina Caffarra (Eurostack Foundation), cited in The Register

(22 December 2025): “A company subject to the extraterritorial laws of the United States cannot be considered sovereign for Europe.”

14 GPU metric reformulation illustrated by concrete cases: Microsoft France (2024, cf. note 10) for the limits of sovereign cloud on European soil; G42/Microsoft agreement (2024), where Abu Dhabi had to agree to sever ties with Chinese entities such as Huawei as a condition of access to advanced NVIDIA chips (US Department of Commerce, statements by Secretary Gina Raimondo); acquisition of Solvinity (Dutch cloud provider) by Kyndryl (American group) in November 2025, de facto cancelling the sovereignty of Dutch public, sector clients who had specifically chosen it (The Register, December 2025).

Key References

International institutions: IEA (2025–2026), World Bank (2025), World Economic Forum (2025–2026), ECLAC/CENIA, ILIA 2025.

Think tanks and research: Bruegel, Brookings Institution, Carnegie Endowment for International Peace, CSIS, Hudson Institute, ITIF, Centre for Future Generations (CFG), Epoch AI.

Consultancies and analysts: McKinsey & Company, S&P Global, Arizton, Gartner, Futurum Group, Deloitte.

Regulatory sources: White House / BIS (AI Diffusion Rule, AI Action Plan, Section 232), European Parliament, European Commission (AI Continent Action Plan, Apply AI Strategy, Chips Act), ANSSI (SecNumCloud).

Specialised press: Euronews, Bloomberg, DCD, Financial Times, Foreign Policy, Pillsbury Law, CM Trade Law, Introl.